

Week 2 Assignment Report

Python for AI & Data Handling

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Internship:	AI Internship
Week:	Week 2
Tools Used:	Python, NumPy, Pandas, Matplotlib, Jupyter Notebook

This report summarizes the work done for Week 2, covering NumPy fundamentals, Pandas dataset handling, data cleaning, data visualization, and a mini project analysing a student performance dataset. Full working code with executed outputs is available in the accompanying Jupyter Notebook (**Week2_Assignment_PythonForAI.ipynb**), and the raw and cleaned datasets are submitted as CSV files alongside this report.

Task 1: NumPy Fundamentals (20 Marks)

NumPy is a Python library used for fast numerical computation. It provides the **array** data structure, which allows mathematical operations to be applied to entire collections of numbers at once, instead of writing manual loops. Almost every AI/ML library in Python (Pandas, Scikit-learn, TensorFlow) is built on top of NumPy, which makes it a foundational tool for this internship.

What was implemented:

- Created 1D and 2D NumPy arrays from Python lists.
- Checked array **shape** and **ndim** (number of dimensions).
- Performed indexing and slicing on both 1D and 2D arrays (e.g. **arr[1:4]**, **arr_2d[0:2, 0:2]**).
- Applied basic mathematical operations (addition, multiplication, squaring) element-wise.
- Used aggregate functions: **mean()**, **max()**, **min()**, **sum()**, and standard deviation, including row-wise and column-wise sums on a 2D array using the **axis** parameter.

Sample Output:

Operation	Result
1D Array	[10 20 30 40 50]
Shape of 2D array (3x3)	(3, 3)
Sum of 1D array	150
Mean of 1D array	30.0
Max / Min of 1D array	50 / 10
Row-wise sum of 2D array	[6 15 24]

Full executed output (all print statements and screenshots-equivalent) is available in **Task 1** of the Jupyter Notebook.

Task 2: Pandas & Dataset Handling (20 Marks)

Pandas is used to load and explore structured/tabular data using a **DataFrame**. For this task, the dataset **student_performance.csv** (63 rows, 7 columns) was loaded and explored.

What was implemented:

- Created a small DataFrame manually from a Python dictionary.
- Loaded the CSV dataset using **pd.read_csv()**.
- Displayed the first 5 and last 5 rows using **.head()** and **.tail()**.
- Checked the number of rows/columns with **.shape** and column names with **.columns**.
- Checked data types of each column with **.dtypes**.
- Generated statistical summary (count, mean, std, min, max, percentiles) using **.describe()**.
- Selected specific columns (e.g. **Math_Marks**) and filtered rows (e.g. students scoring above 80 in Math).

Dataset columns used:

Column	Description
StudentName	Unique identifier for each student
Gender	Male / Female
StudyHours	Average hours studied per day
Attendance	Attendance percentage
Math_Marks / Science_Marks / English	Marks scored out of 100 in each subject

Task 3: Data Cleaning (20 Marks)

The raw dataset was intentionally realistic it contained missing values, duplicate records, and one column with an incorrect data type exactly the kind of issues a real dataset has before it's ready for analysis or Machine Learning.

Before Cleaning:

Check	Result
Total rows	63
Missing values - Gender	2
Missing values - Attendance	4
Missing values - Math_Marks	4
Duplicate rows	3
Attendance column data type	object (text) - because of values like "92%"

Cleaning steps performed:

- **Fixed data type:** Removed "%" from Attendance values and converted the column to numeric using `pd.to_numeric()`.
- **Handled missing values:** Filled missing numeric values (Math_Marks, Attendance) with the column mean, and filled missing Gender values with the mode (most frequent value).
- **Removed duplicates:** Detected with `duplicated()` and removed with `drop_duplicates()`.
- Rounded/converted columns to appropriate final types and saved the result as `student_performance_cleaned.csv`.

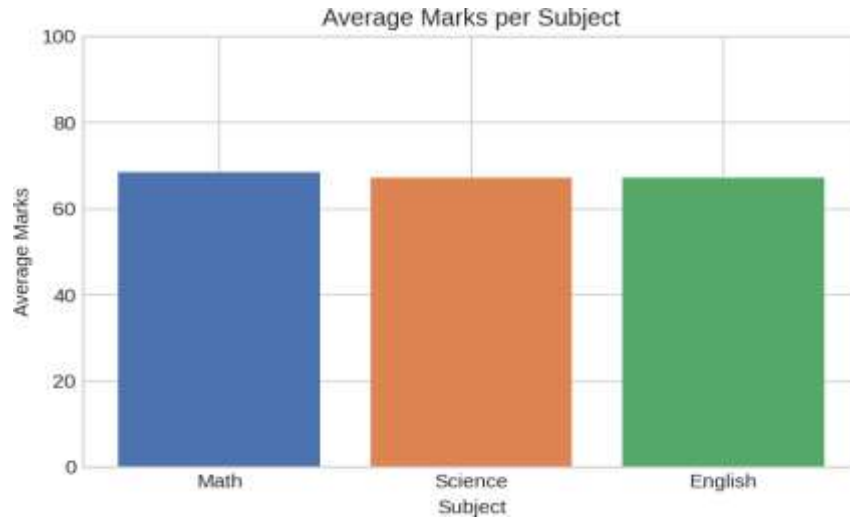
After Cleaning:

Check	Result
Total rows (after removing duplicates)	60
Missing values remaining	0
Duplicate rows remaining	0
Attendance column data type	int64 (numeric) - corrected

Task 4: Data Visualization (20 Marks)

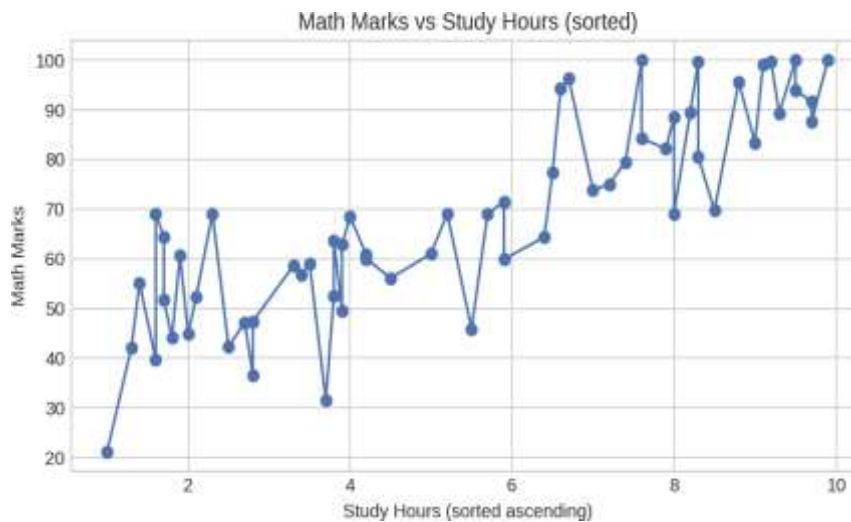
Four different chart types were created using Matplotlib on the cleaned dataset to visually understand patterns in student performance.

1. Bar Chart - Average Marks per Subject



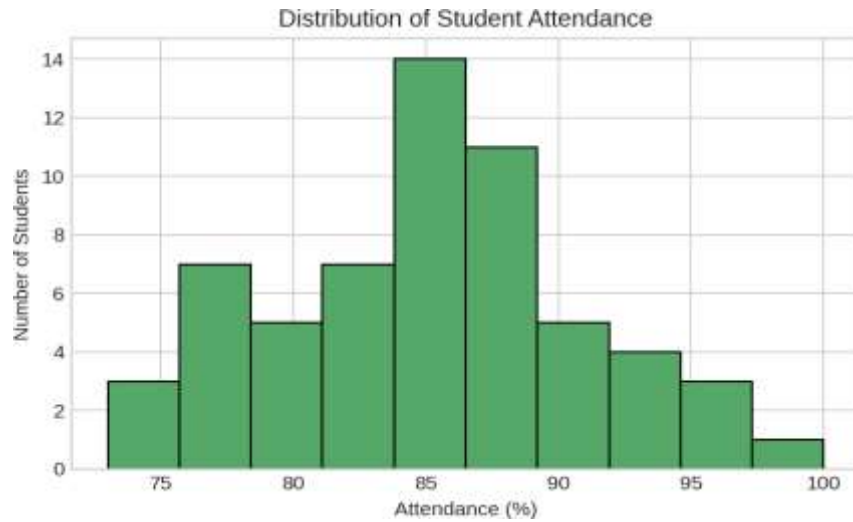
Compares the class average score across Math, Science, and English.

2. Line Chart - Math Marks vs Study Hours



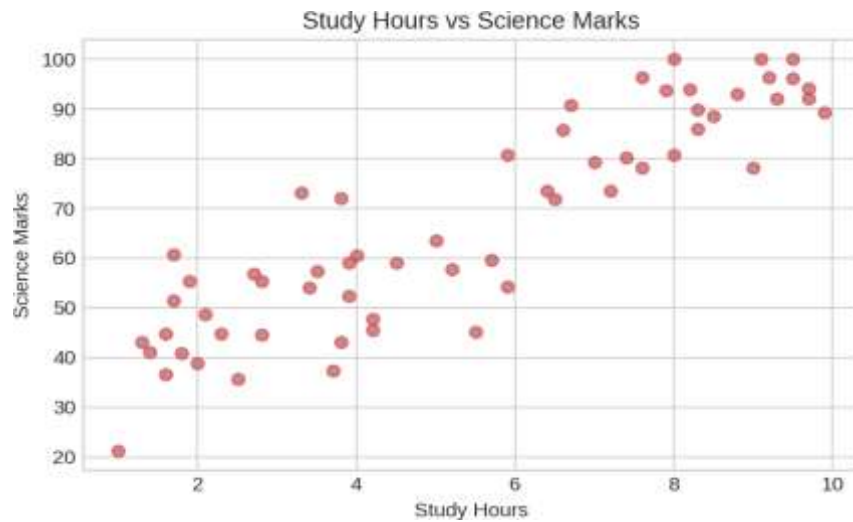
Shows Math marks after sorting students by study hours - an overall upward trend suggests more study hours are linked with higher Math marks.

3. Histogram - Distribution of Attendance



Shows how student attendance percentages are spread across the class.

4. Scatter Plot - Study Hours vs Science Marks



Each point is one student; the upward pattern suggests a positive relationship between study hours and Science marks.

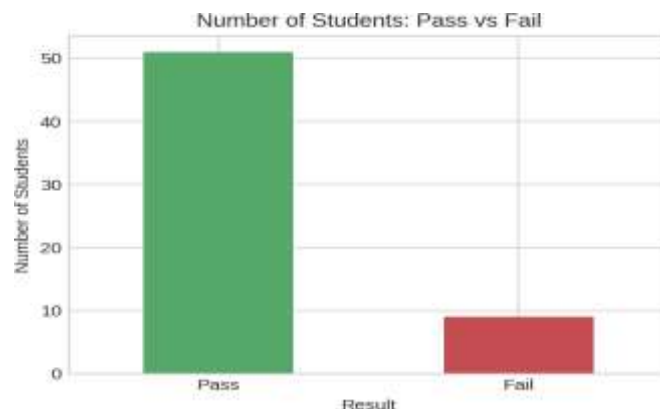
Task 5: Mini Project - Student Performance Analysis (20 Marks)

Using the cleaned dataset, a complete basic analysis of student performance was carried out: computing total/average marks per student, marking each student Pass/Fail (passing mark = 40 in every subject), and summarizing key statistics.

Key Results:

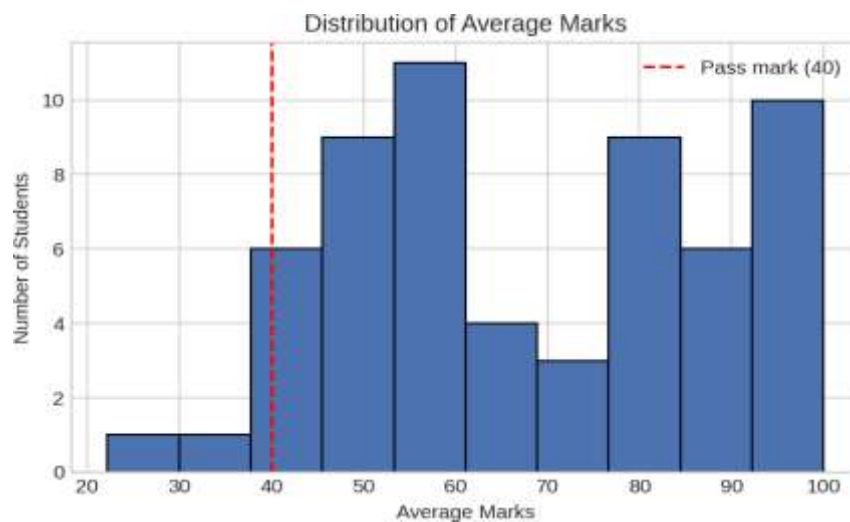
Metric	Value
Total students analysed	60
Overall average marks	67.65 / 100
Highest average marks	100.0 (Student_4)
Lowest average marks	22.2 (Student_43)
Students Passed	51
Students Failed	9
Pass Percentage	85.0%

Chart A - Pass vs Fail Count



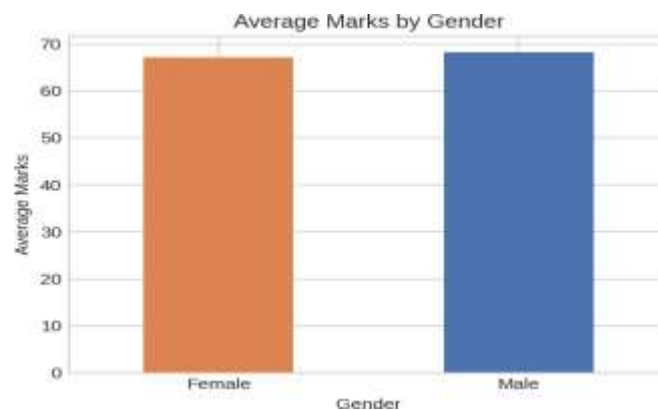
Number of students who passed vs failed based on the 40-mark passing criteria.

Chart B - Distribution of Average Marks



Shows how average marks are spread across the class, relative to the pass mark (40).

Chart C - Average Marks by Gender



Compares average performance between male and female students.

Conclusion

After cleaning and analysing the student performance dataset, a few clear patterns emerge. The overall class average (67.65/100) sits comfortably above the 40-mark passing threshold, and 85% of students passed in all three subjects, with only 9 students falling below the passing mark usually the same students who also reported low study hours and lower attendance. There is a visible positive relationship between **study hours and marks**: students who studied more hours per week tended to score higher, especially in Math and Science, as shown by both the line chart and the scatter plot. Attendance was mostly concentrated in a healthy range, suggesting the class was fairly regular overall, though the few students with lower attendance were also among the lower scorers. When comparing marks by gender, average performance between male and female students was fairly close, with no group dramatically outperforming the other. Overall, this analysis suggests that **study hours and attendance are useful early indicators of student performance**, and could be valuable input features if this dataset were later used to train a Machine Learning model to predict pass/fail outcomes which is the direction this internship is heading in over the coming weeks.

Files Submitted

- Week2_Assignment_PythonForAI.ipynb - full notebook with executed code and outputs (all 5 tasks)

- student_performance.csv - raw dataset (used in Tasks 2 & 3)
- student_performance_cleaned.csv - cleaned dataset (used in Tasks 4 & 5)
- 7 PNG chart images (Task 4 and Mini Project visualizations)
- Week2_Assignment_Report.pdf - this report